

Loans and Credit

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Introduction

This report looks at loan default trends using Lending Club data covering one million anonymised peer-to-peer loans. It looks at the beliefs the Director wishes to challenge. Namely, that home owners and people that's been employed more than 10 years have a far lower probability of default with short term loans, that US States differ significantly when it comes to a culture of defaulting on short term loans and that Credit Grades (A - F) provide a great indicator for whether younger individuals will default or not. Also that interest rates are also clearly determined by age, occupation and credit scores.

Data

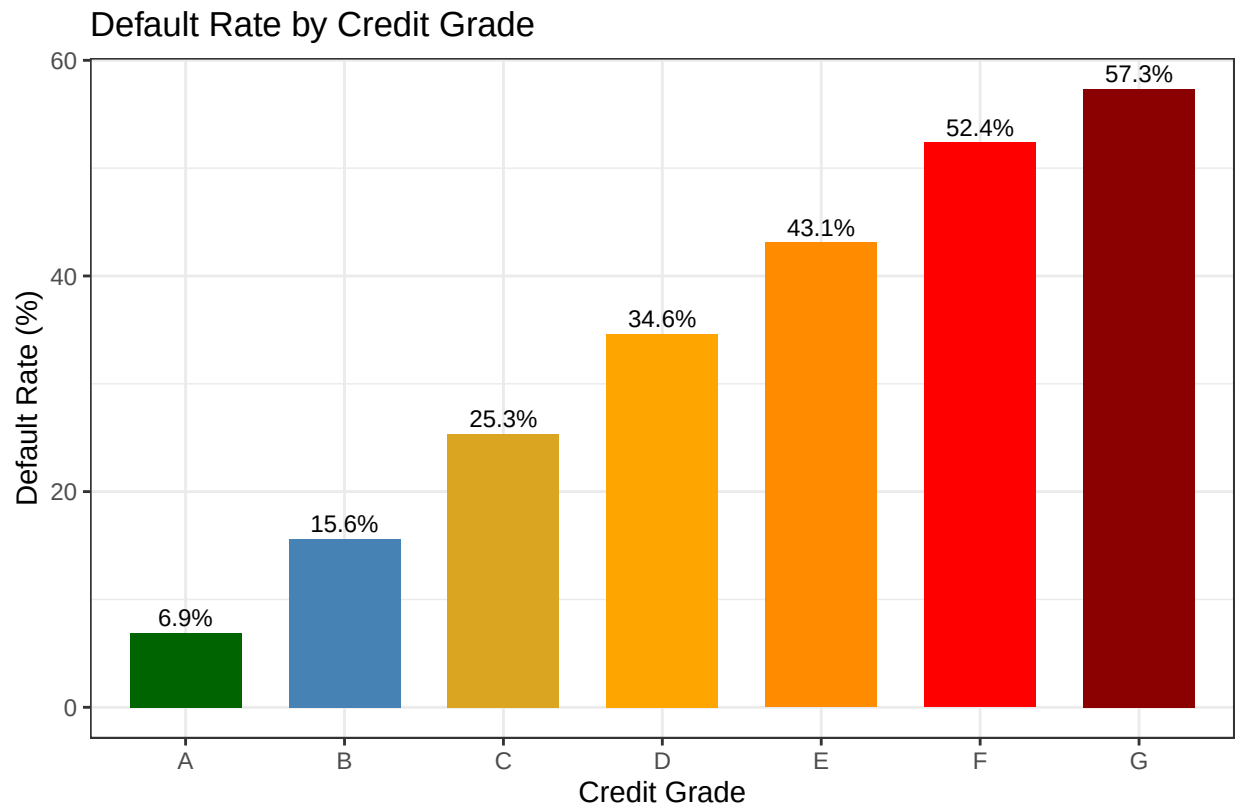
Data Cleaning and Exploration

Before any analysis, the data was reduced to the columns relevant to this analysis and several cleaning steps were applied.

- Two binary outcome variables were created from `loan_status`. `Default_Flag` takes the value 1 for loans charged off or defaulted, 0 for fully paid, and NA for current loans. These are excluded from default rate calculations as their outcome is not yet known.
- Employment length and loan term were stored as text ("6 years", "36 months") and converted to numeric for analysis.
- Credit grade was set as an ordered factor from A (lowest risk) to G (highest risk).
- Credit history length is used as a proxy for age since there is no direct age variable in the dataset. It is derived from `earliest_cr_line` and it is used as a proxy since a longer history is likely linked to an older borrower.
- Continuous variables with clear outliers were Winsorized at the 1st and 99th percentiles. DTI had impossible values ranging from -1 to 999, and annual income reached as high as \$9.9 million so both required capping to prevent extreme values from distorting the results.

Key Risk Drivers

Default Rate by Credit Grade



The plot shows a clear strong relationship between credit grade and default rate (as expected).

Default Rate by Loan Purpose

Car and home improvement loans carry the lowest default rates at around, while small business and wedding loans sit at the top at. Thus, lenders should apply greater scrutiny to higher-risk purpose categories when assessing borrower credibility. This plot provides evidence that loan purpose is a good credit risk signal.

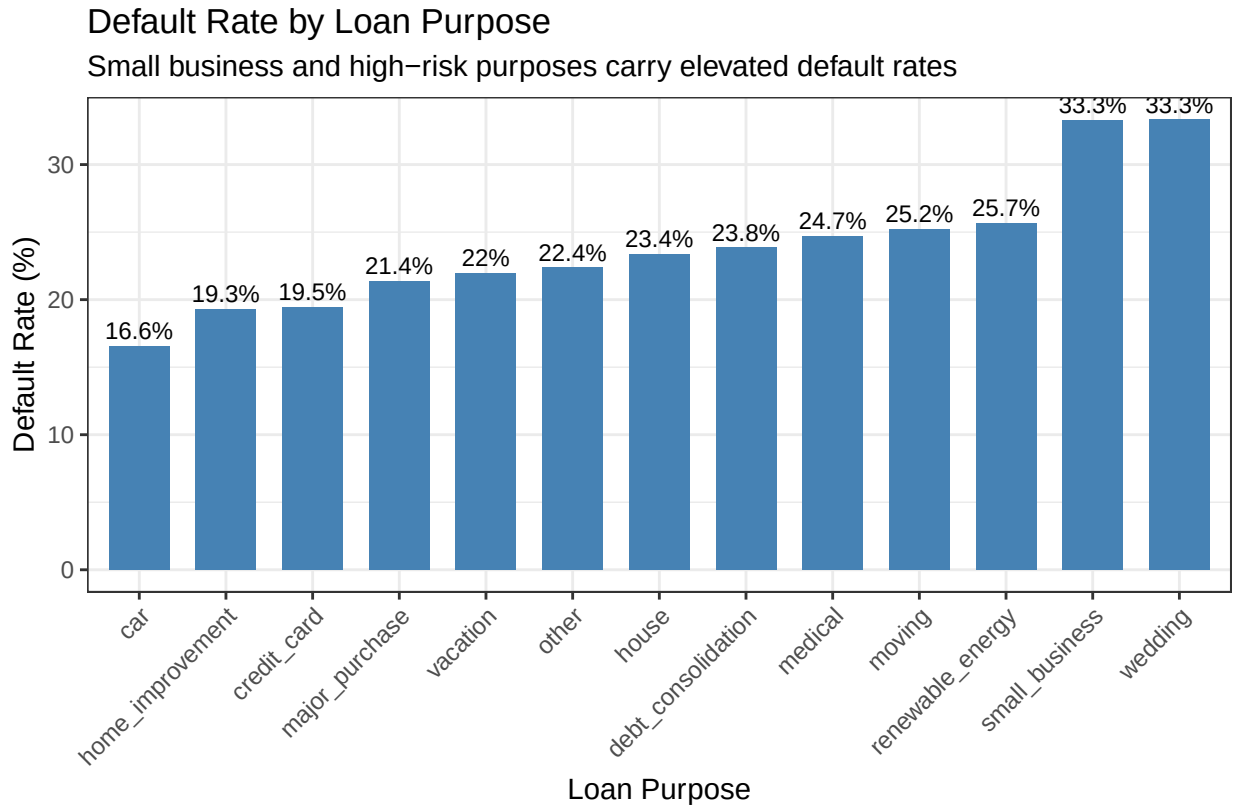
Evaluating the beliefs

Belief: Homeowners and Long-Tenured Employees Default Less

The above plot somewhat confirms this belief since borrowers employed for 10 or more years show slightly lower default rates (lower by 1). This difference is really small so this belief is only partly supported by the data. On the other hand, homeowners (those with over and under 10 years of employment) have higher default rates than borrowers.

Belief: US States Differ Significantly in Default Culture

This belief is more closely supported by the data because default rates range from around 14 to about 27. This means that US states do in fact differ significantly in default culture. Texas (labeled TX on the x-axis)



Source: Lending Club anonymised loan data

Figure 1: Loan purpose varies substantially in associated default risk.

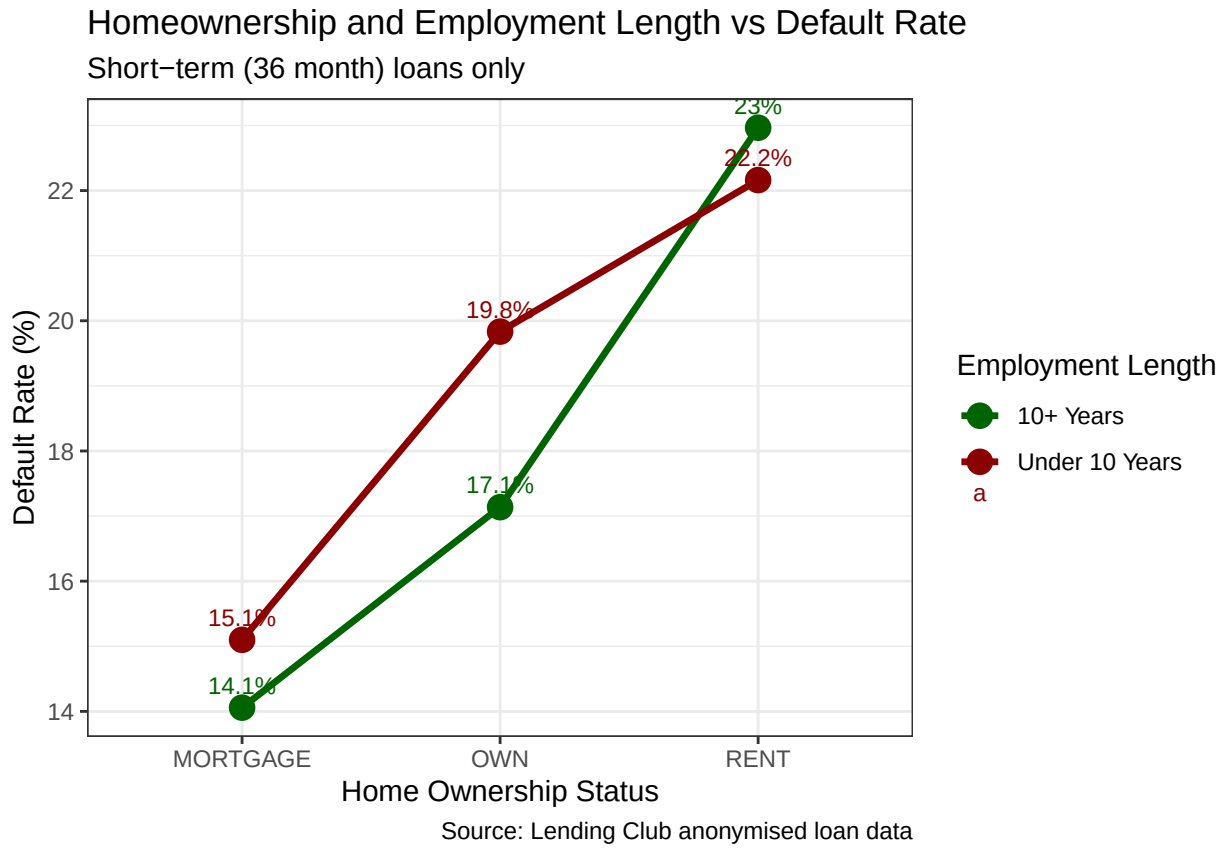


Figure 2: The director's heuristic receives only partial support.

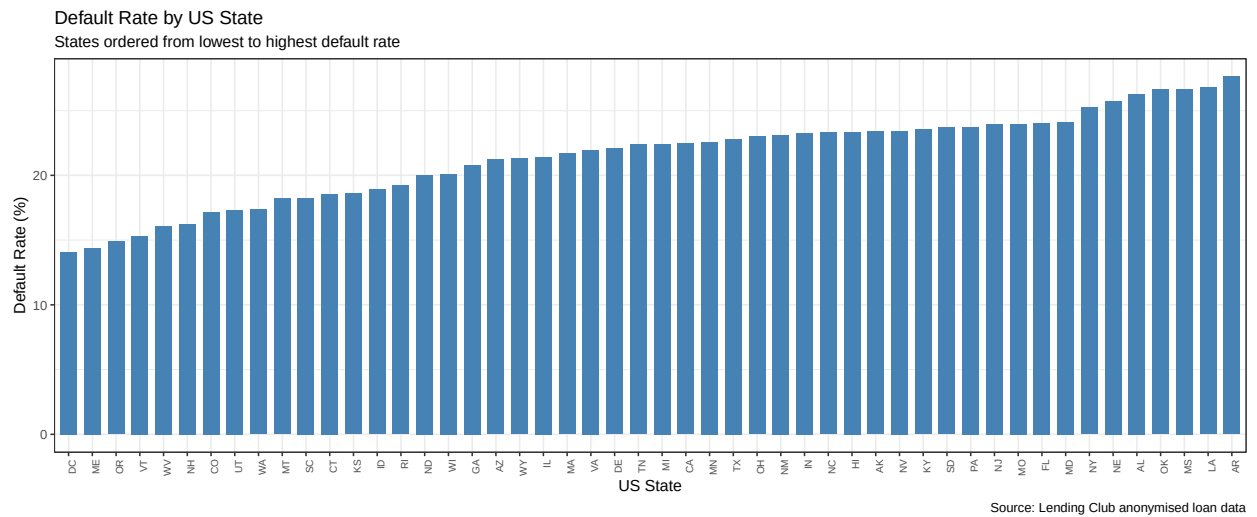


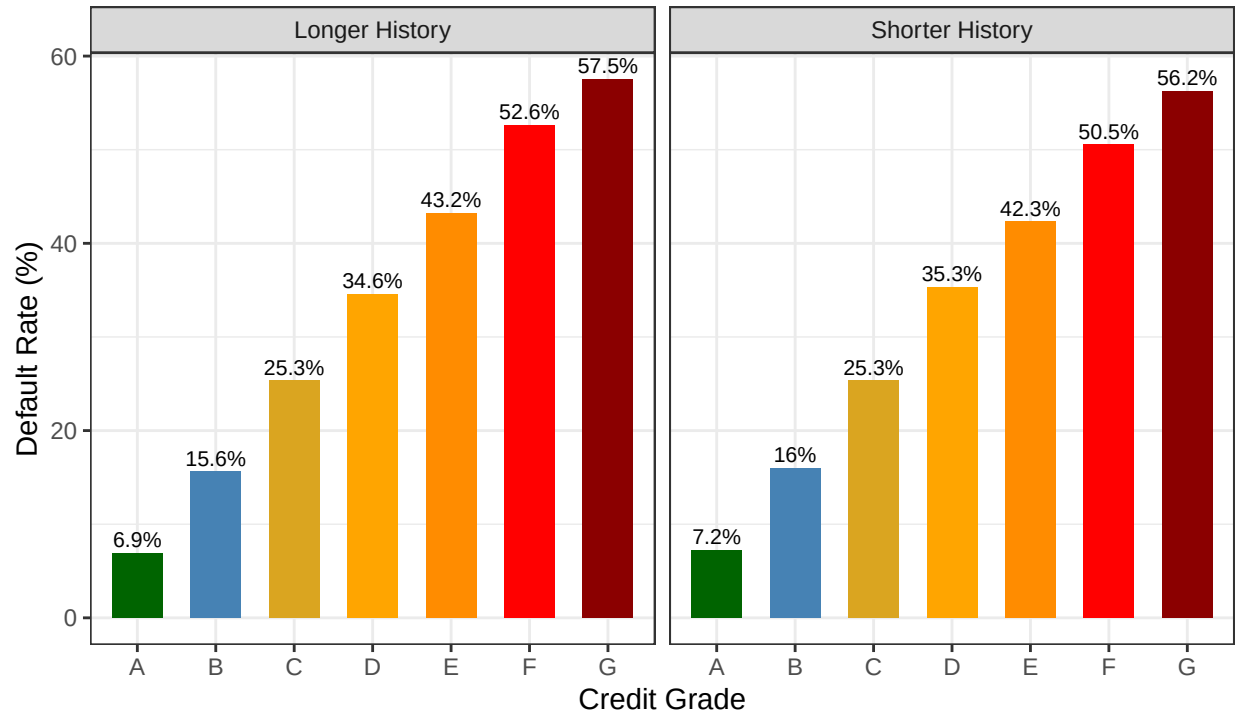
Figure 3: Default rates vary across states but Texas sits close to the national average.

sits slightly above the the average when looking at the bigger picture.

Belief: Credit Grades (A - F) Predict Whether Younger Individuals Will Default Well

Credit Grades vs. Default rates for Older and Younger Individuals

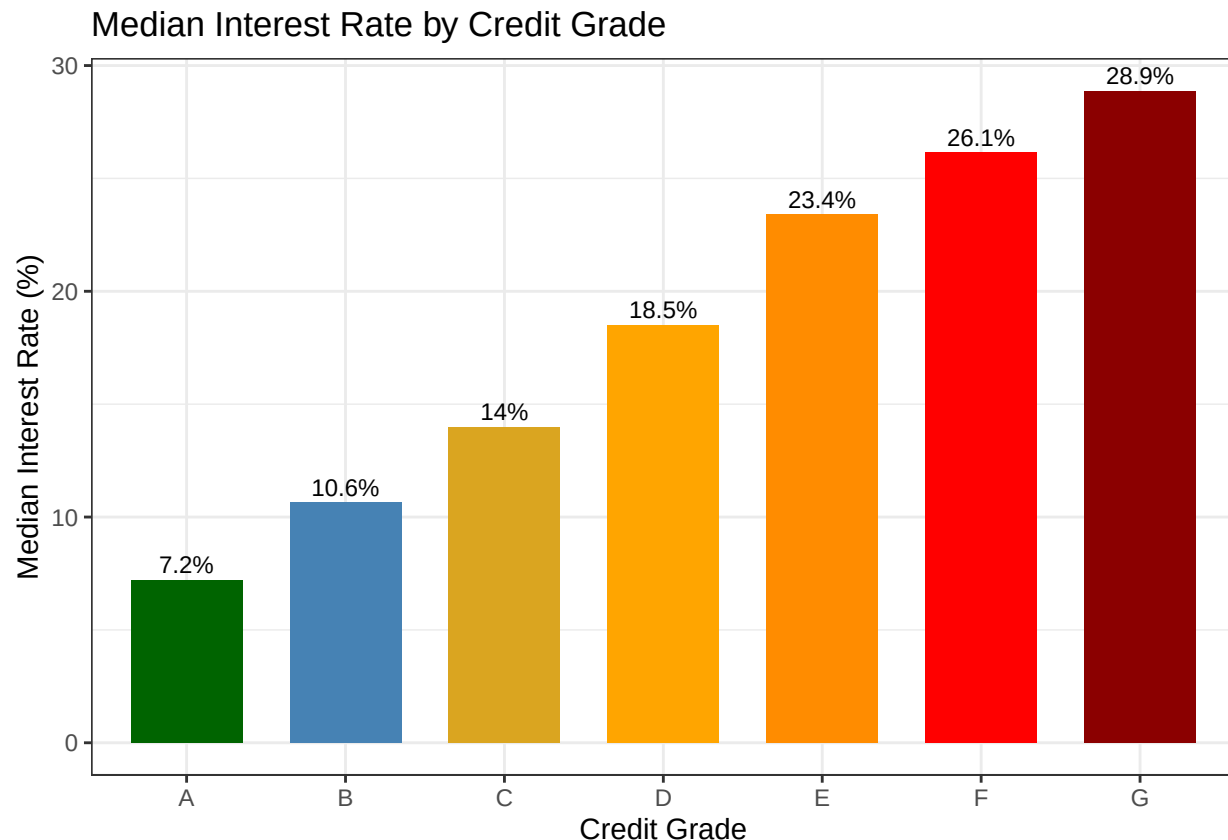
Credit history length used as age proxy



Shorter history = likely younger borrowers

The credit grading system has the same relationship as before for both younger (shorter credit history) and older (longer credit history) individuals. This relationship is that default rates increase with lower credit grades. The differences between older and younger individuals default rates' is very small. For higher grades, younger individuals have a slightly higher default rating while for higher grades it is the opposite. Thus, there is no evidence from the data to support the belief that grades work better or worse for younger borrowers specifically.

Belief: Interest Rates Are Determined by Age, Occupation, and Credit Scores



From this graph it is obvious that credit rating is an obvious determinant of interest rates. From the basic understanding of interest rates and credit ratings, this correlation makes sense, because a lower credit rating means there is a higher risk of default. Thus, the risk is compensated for by a higher interest rate (compared to higher credit ratings).

Conclusion

This report analysed one million Lending Club loans to assess default risk drivers and evaluate four beliefs held by the Credit Institute in Texas.

Credit grade is the strongest predictor of default, with a clear and consistent relationship between grade and default rate. Loan purpose also carries meaningful risk signal car and home improvement loans have the lowest default rates while small business and wedding loans sit at the top.

The evidence on the four beliefs of the Institute is relatively mixed. The homeownership and length of employment belief is only partially supported because borrowers with 10 or more years of employment default slightly less, but the difference is small. The belief that US states differ in default culture is supported, with default rates ranging from around 14% to 27% across states and Texas sits close to the national average. Credit grades predict default consistently for both younger and older borrowers with very small differences between the two groups. There is no evidence that grades work better or worse for younger individuals specifically. Finally, credit grade is clearly the dominant driver of interest rates, but the independent roles of age and occupation cannot be assessed as neither variable is directly available in the dataset.